

Examining Trust Among Veterans from Camp Lejeune Exposed to Environmental Toxicants: A Qualitative Study



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ABSTRACT

BACKGROUND: Serving Veterans with military environmental exposures is a top priority of the Veterans Health Administration (VHA). This priority is critical for over 1 million Veterans and civilians who were exposed to contaminated drinking water at United States Marine Corps (USMC) Base Camp Lejeune (CL) between 1953 and 1987. Since 2012, VHA has enacted various legislation to provide healthcare coverage and monetary relief for harm caused by water contamination.

OBJECTIVE: To understand the perceived impact of toxicant exposure at CL on Veterans' trust in the VHA.

DESIGN: This is a qualitative study based on semi-structured telephone interviews between February 2024 and May 2024 to understand Veteran experiences with toxicant exposure and feelings of trust.

PARTICIPANTS: Twenty-two Veterans who served at CL during the officially recognized contamination period were interviewed (68.2% male; mean age 63 years; 72.7% White).

APPROACH: Recruitment included contacting participants who completed a previous survey and posting an invitation on two private CL Facebook pages. Interviews were audio-recorded and transcribed, then a codebook was developed inductively. During thematic analysis, we synthesized overarching themes, later organized into the Health Systems Trust Content Area Framework.

KEY RESULTS: We developed three key themes. First, reasons for trust included Veterans' positive experiences with VHA healthcare. Second, reasons for lack of trust in Veterans Affairs included poor experiences with healthcare, lack of transparency, and complicated claims processes. Third, Veterans' trust in the VHA was complicated by external factors, including the USMC and US government.

CONCLUSIONS: These findings suggest that CL Veterans' trust in the VHA is shaped by complex factors, including potentially modifiable factors (e.g., healthcare access, transparency). Future efforts to measure

and rebuild trust in this vulnerable population should include continued collaboration with Veterans, drawing on their specific experiences and lessons from past contaminations, for improved health policy and greater VHA service use.

KEY WORDS: trust in healthcare; military toxicant exposures; Veterans health

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INTRODUCTION

Trust in healthcare systems, and the providers, structures, and care processes that underpin them, is essential to deliver timely, high-quality care.^{1–3} Defined as the acceptance of a vulnerable situation where a patient is confident their healthcare provider will act in their best interests,^{3,4} trust permeates all aspects of the care continuum. Increased trust enhances patient willingness to seek medical treatment, disclose information, and follow medical recommendations, and contributes to greater provider willingness to share decision-making authority with their patients.^{1,2} Still, mistrust is prevalent in healthcare settings, contributing to disproportionate harm, particularly among marginalized communities.^{5,6} Recent recommendations highlight the need to effectively measure and maintain trust in healthcare, acknowledge past experiences of marginalized groups, involve communities in decision-making, and ensure transparency and integrity in health practices and policies.^{7,8} To ensure the highest quality of care, dedicated efforts are needed from a healthcare system and policy standpoint to restore trust if it has been eroded.^{9,10}

These recommendations are especially relevant to United States (US) Veterans exposed to environmental toxicants during military service. Between 1953 and 1987, over 1 million Veterans and civilians were exposed to toxicants in the drinking water at the United States Marine Corps (USMC) Base Camp Lejeune (CL), in Jacksonville, North Carolina. The water was contaminated with industrial solvents from a nearby dry-cleaning facility and benzenes,

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ubiquitous environmental pollutants recognized by the International Agency for Research on Cancer as human carcinogens,¹¹ from underground fuel storage tanks leaking into the base's water source.¹² Decades later, many exposed residents continue to face serious health conditions presumed to be related to their exposure.¹³ Addressing the healthcare needs of Veterans with military environmental exposures continues to be a key priority for the Veterans Health Administration (VHA),^{14,15} one of three administrations within the Department of Veterans Affairs (VA), and underpins several legislative efforts enacted in this area.^{16–18}

The impact of environmental exposures often extends beyond health, influencing an individual's perceptions and experiences of healthcare, potentially due to feelings of institutional betrayal: situations in which the institutions people depend upon for safety and well-being, including the US government, cause them harm.^{19,20} Institutional betrayal may trickle down to other government-related institutions, such as the VA and the VHA, impacting trust in healthcare.²⁰ While measuring trust among Veterans has become an established practice in the VHA,²¹ our understanding of trust in the VHA among Veterans with military environmental exposure at CL remains limited. Veterans receiving care from the VHA report high trust in the institution, while Veterans who choose not to or cannot receive care from the VHA often report lower levels of trust.²² Low levels of trust are associated with poorer health habits, delayed checkups, more missed appointments, and overall unmet medical needs.¹ Understanding the trust of Veterans with military environmental exposure is critical as it may impact whether Veterans seek life-saving healthcare. As part of a broader national policy evaluation, we conducted a qualitative study to understand how CL Veterans' military environmental exposure has impacted their feelings of trust in the VHA.

METHODS

Context: Background and Scope of This Work

Congress and the VHA enacted several pieces of legislation which provided CL Veterans who served during the officially recognized contamination period with healthcare and monetary relief for associated health effects. The 2012 Honoring Veterans and Caring for Camp Lejeune Families Act (PL 112–154) provided free VHA care for Veterans and out-of-pocket reimbursement for family members who were at CL for at least 30 days and diagnosed with any of the 15 associated conditions (Appendix 1).¹⁷ Having one of the eight presumptive conditions (Appendix 1) enabled Veterans to seek disability compensation, as these conditions were automatically assumed to be caused by exposure at CL.¹⁸ The CL Justice Act of 2022 is a provision and Title (VII) of the Promise to Address Comprehensive Toxics (PACT) Act which provided

Veterans and affected others the opportunity to submit a one-time claim between August 2022 and August 2024 for harm due to the toxicants in the water if they (or their mother while pregnant with them) were exposed at CL for at least 30 consecutive days.¹⁶ The PACT Act extends beyond CL Veterans to other Veterans with toxic exposures and Veterans of the Gulf War and Post-9/11 eras; therefore, the exploration of trust with CL Veterans, who are now between 57 and 90 years old, may have relevance for other groups of Veterans more recently impacted by environmental toxicants and may face similar trust issues with their care. In 2023, the Evidence to Practice Evaluation Center for Veterans (EPEC-Vet) was funded to assess the impact of military environmental toxicant legislation on Veterans' access to and use of exposure-related VHA services, including this study on CL Veterans.

Qualitative Data Collection

Previously, our team conducted a survey to assess the impact of the CL related legislation on Veterans' experiences with health care access and disability claims.²³ Veterans who completed this survey ($n = 1073$) were recruited from (a) a 10% stratified random sample of Veterans based at CL between 1975 and 1985 with available mailing addresses as provided by the VA Corporate Data Warehouse (with oversampling of underrepresented groups, including women, Veterans of non-White race, and Hispanic Veterans), or (b) snowball sampling where a survey invitation was posted on two private/monitored Camp Lejeune Facebook sites.²³ Veterans who had been at CL for at least 30 consecutive days during the officially recognized contamination period and completed this previous survey conducted by our research team could volunteer for an in-depth telephone interview regarding their experiences. An invitation was again also posted on two private CL Facebook sites to identify additional participants who may not use VA care. We developed an interview guide by assessing patterns and gaps in previous survey results.²³ Questions included (1) Veterans' awareness, perceptions, and experiences related to toxicants at CL, (2) corresponding health issues and healthcare, and (3) how these experiences related to toxicants at CL affected their trust in the VHA. We conducted pre-tests of the interview guide by interviewing two Veterans based at CL during the contamination period; no major edits were made, and these two interviews were used as part of our final analysis.

We conducted interviews in pairs of qualitative team members (FW, SB, CI, KB, PS, JB). The lead interviewer asked questions, while the other took notes and asked follow-up questions. All team members were female and worked as researchers at the VA and/or a university. Interviewers had training in qualitative research and extensive experience with interview facilitation. Participants had no pre-established relationships with the interviewers. Veterans were told the goal of the interview was to understand their experiences related to the CL legislation and how

best to support them. We collected demographic information when interviews were scheduled. All interviews were conducted, audio-recorded, and transcribed verbatim using Microsoft Teams, then reviewed for accuracy. No repeat interviews were conducted. Veterans received a \$20 gift card for their participation.

We reported this qualitative study according to the Consolidated Criteria for Reporting Qualitative Research (COREQ) checklist.²⁴ The VA's Office of Research and Development determined that this evaluation is classified as quality improvement and did not meet the regulatory definition of research in accordance with 38 CFR 16.102(l). While approval was not required from the Institutional Review Board, verbal informed consent was obtained from all participants.

HSTCA Framework

The Health Systems Trust Content Area (HSTCA) Framework is based on a systematic literature review of

measurements of trust within the health system: competence, honesty, communication, confidence, fidelity, fairness, confidentiality, and systems trust.²⁵ Prior literature has extensively investigated key factors related to trust and has defined these constructs. To inform our analytic approach, we adapted the definitions of the eight domains, originally developed to measure trust in the general health system,²⁵ to create operational definitions pertaining to CL Veterans during and after service as described in Figure 1.

Thematic Analysis

A codebook was developed inductively. Three researchers coded transcripts to identify patterns and synthesize overarching themes using Dedoose Desktop Version 9.2.12. The study team determined saturation was reached when no new themes emerged in the coding process.²⁶ We then organized themes according to the eight domains of the HSTCA Framework as previously described. By using a trust framework to analyze Veteran experiences, we aim

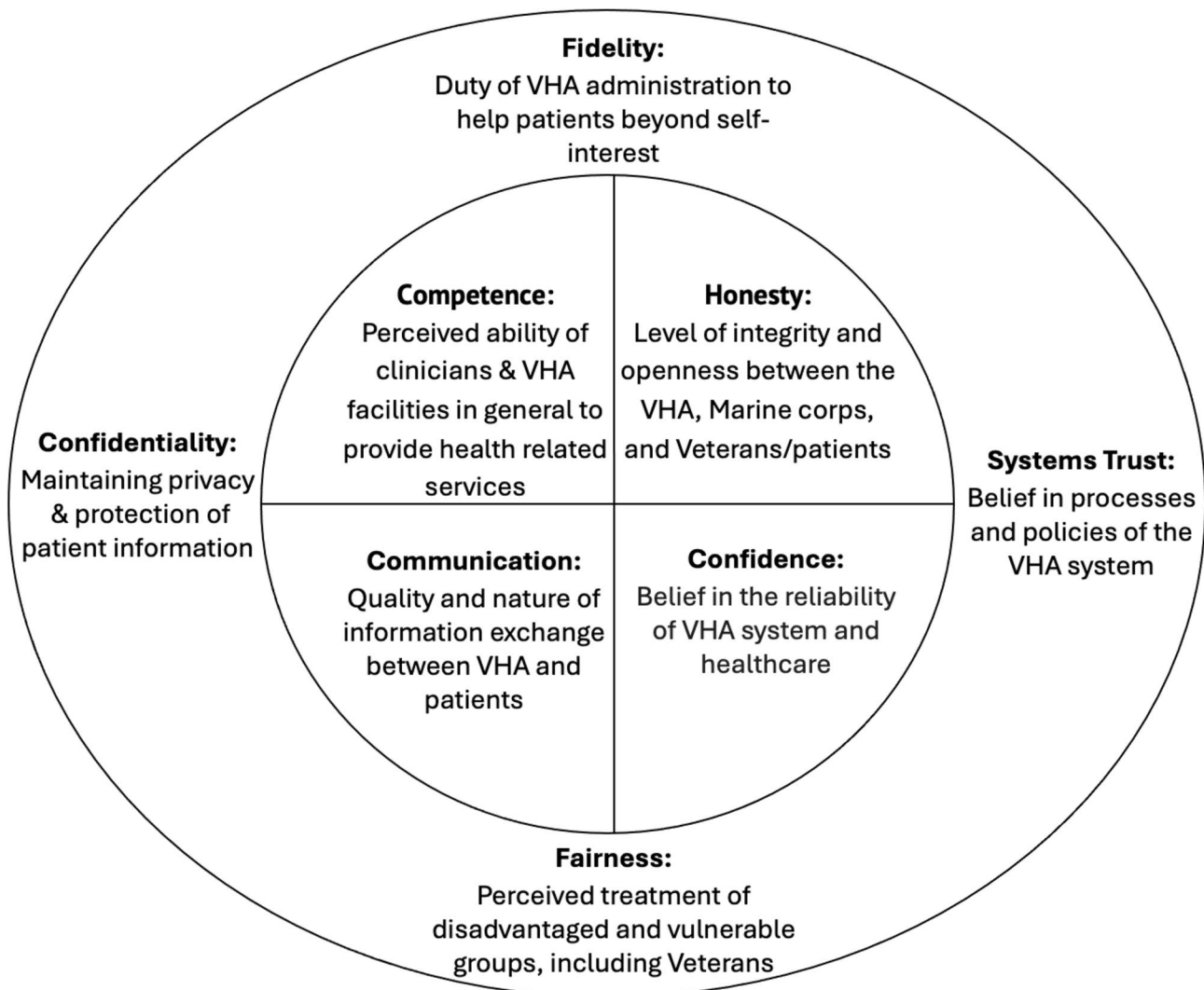


Figure 1 Health Systems Trust Content Area Framework with operational definitions (adapted from Ozawa & Sripad).²⁵

to identify standardized areas of trust established in the literature that are critical to the response at CL and can also be applied to future military environmental exposures.

RESULTS

We conducted interviews between February and May of 2024 using a semi-structured guide which lasted approximately 30–45 min. Of the 24 volunteers from the past survey, 15 were reached and completed interviews, followed by an additional 7 volunteers from Facebook, for a total of 22 interviews. A majority of participants were male

($n = 15$) and White ($n = 16$) with a mean age of 63 years. Participant characteristics are presented in Table 1.

Overall, participants described a wide range of experiences with the VHA, associated legislation, and claims processes. We developed three key themes: (1) *trust* due to Veterans' positive experiences with VHA services and perceived transparency about toxicant exposure, (2) *lack of trust* due to limited access to or poor experiences with healthcare, lack of transparency, and complicated claims processes, and (3) *external influencers impacting trust* shaped by perceptions of the USMC and US government. A summary of how each HSTCA domain connects to each theme can be found in Table 2. Many Veterans expressed a combination of these three themes, revealing the complexity of trust in the VHA. Each illustrative quote corresponds to an HSTCA domain denoted in underlined text and includes a description of Veteran sex and years served at Camp Lejeune, for example (Male, 1978–1981); illustrative quotes are showcased throughout the results section with additional quotes in Table 3.

Table 1 Characteristics of Camp Lejeune Interview Respondents

Characteristics	Participants ($n = 22$)
Age, mean (SD)	63 (5.30)
*Prefer not to answer/missing	$n = 1$
Gender	
Male	68.2% (15)
Female	31.8% (7)
Race	
White	72.7% (16)
Black or African American	22.7% (5)
Asian	4.6% (1)
Ethnicity	
Non-Hispanic or Latino	100% (22)

Theme 1: *Trust* due to Veterans' Positive Experiences with VHA Services and Perceived Transparency About Toxicant Exposure

Within the competence domain, defined as the perceived ability of clinicians and VHA facilities to provide health

Table 2 Summary of Key Themes Organized by HSTCA Domain

HSTCA domain	Theme 1: Trust	Theme 2: Lack of trust	Theme 3: External influences on trust
Communication	Providers were kind and enjoyable during patient appointments	Dissatisfaction with the clarity of what was causing symptoms, how disability works, what was covered, and the claims processes	Frustration with how the government shared information relating to the contamination and associated difficulties with navigating this information
Competence	Satisfaction with available services, continuous care, early detection, & clinician expertise	Inadequate care for those affected, including a lack of correct or timely diagnosis and treatment	The government's handling of other events impacted the perceived ability of the VHA to provide health-related services
Confidence	Reliability of the VHA to provide necessary care and information, including answering questions	Lack of reliability of VHA clinicians	Lack of reliability of the government even though Veterans served their country and did their part
Confidentiality	–	–	Perceived failure to protect information through records that were believed to be lost or burned
Fairness	–	Belief that everyone exposed regardless of time and dependent status should be treated equally, but some Veterans did not feel valued in the decision-making process and felt their symptoms were dismissed	Veterans described how they did not agree to be exposed upon signing up for service and how dependents were exposed without being a service member
Fidelity	Felt the VHA was fulfilling their duty to financially take care of Veterans	Felt VHA had limited interest in truly providing CL Veterans with associated compensation or care	Frustration that Veterans were exposed to environmental toxicants under government supervision, eroding existing trust in the government
Honesty	Open with Veterans about exposure to environmental toxicants by mailing a letter	Perceived lack of transparency from the VHA about the water contamination issues	Perceived lack of transparency from the USMC and US government, rather than the VHA
Systems trust	Belief that the policies and claims process would support Veterans without much difficulty	Multiple claim denials led Veterans to question functionality of claims process	–

Table 3 Illustrative Quotes Linked to HSTCA Domains

HSTCA Domain	Illustrative Quote (Gender, Years at CL)
Theme 1: <i>Trust</i> due to Veterans' positive experiences with VHA services and perceived transparency about toxicant exposure	
Competence	"VA has been top notch...I went for an unrelated problem, she suggested I get a urine test that showed that I had bladder cancer. Ever since then, they've been on top of it, making sure things [are] taken care of." (Male, 1980a)
Confidence	"That helped with my trust, that they would take care of something if something is really bothering me." (Male, 1982–1984)
Communication	"I've kinda grown fond of the people that I go to see. I get along with them really well. I actually had one of the doctors write that I'm the best patient she's ever had." (Male, 1984–1987)
Honesty	"I'm very confident with the medical facility here and the openness of the study that was done, how it's been provided to me, and how the VA has grasped that situation and is now caring for me." (Male, 1978–1981)
Systems trust	"The VA I have never had a problem with. It requires to get the paper-work exactly right, it's gonna get worked in. I just hadn't been able to do that yet. So I have no problem with the VA." (Male, 1980–unknown)
Fidelity	"I have a high level of trust in the VA since they were so quick in dealing with all the documentation that I submitted and getting me disability... the VA has gotten me out of the financial hole that was in." (Male, 1960–1964)
Theme 2: <i>Lack of trust</i> due to limited access to or poor experiences with healthcare, lack of transparency, and complicated claims processes	
Competence	"I don't trust 'em...they said make sure you go to the doctor if you saw something, I went, they checked it and said everything was all right. Next thing you know, I'm still havin' trouble." (Male, 1978–1981)
Fidelity	"I don't think the VA is that nefarious, but the more people that give up, the better off they are. It's like 'Ignore Agent Orange for 60 years, and they'll all be dead.'" (Male, 1981–1984)
Honesty	"We had no idea when we got there or left what was goin' on, and it took this long for them to come out and say, yep. Hey, something is going on here. This is what we're doing." (Male, 1984–1987)
Communication	"You are the first person in the VA system, besides all the active Veterans Affairs officers, that has listened to me talk in detail about multiple myeloma and what I've gone through...they get a big F on [trust]." (Female, 1979, 1981)
Systems trust	"All the information, the legislation saying that they're gonna take care of the Veterans... it's lip service. They're saying they're gonna do somethin', and really, they're not 'cause the paperwork that we gotta go through is ridiculous." (Female, 1978–1982)
Confidence	"It took 'em almost two years to get me in to try to do the biopsy and then they canceled it." (Male, 1984b)
Fairness	"I spent a lot of time in the Marine Corps... I've watched a lot of 'em die, and a lot of 'em get totally screwed up. It's not fair. They need to do something before these guys and gals die." (Male, 1984–1987)
Theme 3: <i>External influencers impacting trust</i> shaped by perceptions of the USMC and US government	
Competence	"Unfortunately, the VA gets under the umbrella of government, and so when you're under that umbrella, my opinion can be a little distrustful. And I think we saw that with COVID and the vaccine issues." (Female, 1985)
Fidelity	"Before this event, my husband and I are very pro-military. We're what we call lifers...but when I read about what they've done to people who are affected by things that the government did, that bothers me that they're willing to do that to thousands and thousands of people who have served this country." (Female, 1985–1987)
Honesty	"I'm mad at the Marines because they knew about it and continued to let this happen...People on the top weren't doin' their job and it's the same way it probably always has been. We don't count, we don't matter." (Female, 1984–1990)
Communication	"It helped me to understand just how backwards the government/VA is. Their paperwork, their hardware, their software, their communication skills. All of that is not worth a poop. Not worth a darn. It scares the heck out of me to even think that some day I will be going to the VA and using the benefits that I, as a Marine, am entitled to." (Female, 1978–1982)
Confidence	"They're not somebody you can count on. I guess if you're putting it in a trust way, you know if you were thinking they'll take care of me because I did my thing for them. No, not looking that way." (Female, 1982–1985)

Table 3 (continued)

HSTCA Domain	Illustrative Quote (Gender, Years at CL)
Confidentiality	“They said that my records along with thousands of other records were all burned. They had a fire or something...I tried to get my brother’s records. His records were gone too, so something is going on.” (Male, 1984–1987)
Fairness	“...my trust in the government is not quite what it could be...you joined with the understanding that you’re going to serve your country, but it really was something when I found out that they actually did have contaminated water. That’s not what I signed up for.” (Male, 1980a)

related services, Veterans described satisfaction with the VHA’s capacity to provide adequate healthcare services and continuous care, the expertise of VA clinicians, and their use of screening tests. As one Veteran stated,

“When I got COVID in 2020, I wound up getting pneumonia... They gave me some medicine and in a week or two, got better. So that’s another thing that helped build my trust.” (Male, 1982–1984)

The confidence domain, defined as the belief in the reliability of VHA system and healthcare, revealed Veterans’ certainty that the VHA would provide necessary care and information: “I know personally that I can reach out to the VA at any time and ask these questions” (Male, 1978–1981). Additionally, Veterans expressed their satisfaction with the VHA’s communication (i.e., the quality and nature of information exchange between VHA and patients), noting that clinicians were kind and enjoyable during appointments: “They were awesome. They have been the whole time we’ve been here, goin’ on three years. They’ve just been friendly, and they know your name when you walk in” (Male, 1984–1987). Some Veterans also felt that the VA showed honesty, defined as the level of integrity and openness between the VHA, Marine corps, and Veterans/patients, by being open about Veterans’ military environmental exposure at CL, typically alerting Veterans of the water contamination through a mailed letter.

The systems trust domain, defined as the belief in processes and policies of the VHA system, focused on the claims process; some Veterans described that if claims paperwork was completed correctly, they felt assured that the VA would adequately support them. Within the fidelity domain, defined as the duty of VHA administration to help patients, Veterans generally spoke about insurance and medical bill coverage; they believed that the VA was fulfilling their duty to financially support Veterans by providing access to healthcare, monetary relief, and expensive repeat tests. One Veteran stated,

“If it wasn’t for the VA, my cancer probably would not have been diagnosed until it was in later stages. The VA provided care that would not have been routinely covered by my insurance company if I went to an independent or a private practice...there was no question or no maneuvering or no insurance hoops to jump through.” (Male, 1981–1984)

Respondents did not share any experiences related to reasons for trust that met the HSTCA’s description of confidentiality (maintaining privacy and protection of patient information) or fairness (perceived treatment of disadvantaged and vulnerable groups, including Veterans).

Theme 2: Lack of Trust due to Limited Access to or Poor Experiences with Healthcare, Lack of Transparency, and Complicated Claims Processes

Competence typically described Veterans’ perceptions that clinicians provided inadequate care for those affected, including a perceived lack of correct or timely diagnosis and treatment. Many believed this was due to a fast turnover of clinicians at VHA facilities, resulting in more clinicians inexperienced with working with Veterans and the VHA system:

“In the last five years, I’ve had five different primary care doctors at the VA ‘cause they don’t stick around very long, so they don’t get to know their patients... when people don’t stick around, doctors, you kinda begin to wonder.” (Male, 1984a)

Within this theme, fidelity referred to some Veterans’ belief that the VA had limited interest in providing CL Veterans with compensation or care. For example, some described difficulty receiving appointments which they believed was because they were from CL. They believed that the length of the claims process and the denial of claims indicated that the VA prioritized cost savings over providing affected Veterans with necessary treatment and compensation. One Veteran shared his belief that the VA has not updated the presumptive list because they know this would “open the doors” for more Veterans to get treatment:

“A lot of our Veterans are not getting any healthcare and a lot of ‘em are on the street because of such high denial rate. We shouldn’t have any Veterans living on the street, none...the denial rate has got to change.” (Male, 1984b)

The honesty domain referred to the perceived lack of transparency related to the CL water contamination, and feelings that the VA was being secretive. Frustration with the lack of information provided about the contamination and the amount

of time it took the VA to provide subsequent benefits led to beliefs that the VA was not being honest. One Veteran stated,

“They took 20 years to tell us about this, and we get two years to attempt to get something back from it, and all we get is BS bureaucratic, we need this paper, we need that paper, and we get no communication. It’s not right.” (Female, 1978–1982)

This related to communication as Veterans expressed a lack of clear communication about what might be causing their symptoms, how disability worked, what was covered, and how their claims were progressing: “Putting in the disability claims is just, hurry up and wait. And you just sit there and wait, and wait, and wait, and wait, and wait. Still haven’t heard anything” (Female, 1982–1985).

Systems trust captured Veterans’ experiences receiving multiple disability claim denials, leading to the belief that even if they completed the paperwork, they would still not be approved. Veterans often expressed frustration with the claims process because of its perceived lack of coverage for services used to manage early warning signs and symptoms, feeling that their conditions must seriously progress before services would apply to them: “Just to think about it, it upsets me. I have to be dyin’ before you care about me...you almost have to go through the red tape to get taken care of. I don’t like that” (Male, 1983). The confidence domain related to Veterans who felt unable to depend on healthcare providers for the care they felt they needed, often mentioning differences between VA locations:

“We’re losing Veterans every day that were not getting the treatment, the benefit, that they deserve from being at Camp Lejeune...it seems like each individual hospital or clinic does whatever they wanna do and nobody’s overseeing what’s really goin’ on. They’re screwing us all over.” (Male, 1984b)

Veterans shared experiences of feeling disrespected during healthcare appointments and felt their symptoms were dismissed; this led to a perception of a lack of fairness as some Veterans did not feel valued in the decision-making process. Outside of the healthcare received, a few Veterans discussed that everyone exposed, including those at CL for less than 30 days and dependents, should be compensated and treated the same as those who were at CL for over 30 days: “All of us there for two weeks or four weeks deserve [treatment/compensation]...you guys are keeping us out so you don’t have to be responsible” (Male, 1980b).

No participants described experiences related to reasons for lack of trust that met the criteria for the confidentiality domain.

Theme 3: External Influencers Impacting Trust Shaped by Perceptions of the USMC and US Government

The competence domain described how actions by the US government (e.g., response to the COVID-19 pandemic)

shaped how Veterans perceived the VHA’s ability to provide health-related services. For the fidelity domain, Veterans expressed frustration that thousands of military members were exposed to environmental toxicants under supervision of the US government, thus not fulfilling their duty to help Veterans, often eroding their existing trust in and support for the government. Within the honesty domain, Veterans again perceived a lack of transparency, however, this time their frustrations were directed towards the USMC and US government, rather than the VA. They reported feeling as though these institutions were trying to hide the contamination because of how long it took them to provide Veterans information: “As far as the trust of the government, when that water contamination was first discovered, they tried to sweep it under the rug and hide it” (Male, 1984a).

In terms of communication, there was frustration with how the government shared information related to the contamination and associated difficulties with navigating this information, often involving heavy legal and legislative jargon; although unclear when each Veteran tried accessing information, several Veterans noted that links to CL-related information on government websites were broken, phone numbers were out-of-date, and webpages were reportedly down during updates. The confidence domain included the expressed lack of consistency in the provision of services by the government even though Veterans served their country and did their part. Confidentiality was related to Veterans’ perceived barriers to accessing records needed for disability approval. Veterans reported that their records were lost or burned in a fire,²⁷ believing that the US government and USMC failed to protect their information. One Veteran shared,

“I was denied because they lost my active-duty records. My reserve records are not doing any good for getting me claims...that’s what makes me upset.” (Male, 1983)

Fairness related to the perceived treatment of Veterans who were unknowingly exposed to contamination while serving their country. Veterans described how they did not agree to be exposed when they signed up for military service and the impact of the contamination on their dependents at CL who were unfairly impacted by contaminated water at a government-run military base.

No described experiences related to external influences on trust met the criteria for the systems trust domain.

DISCUSSION

Veterans impacted by military environmental exposures at CL reported mixed feelings regarding their trust in the VHA. Veterans had a combination of trust, lack of trust, and trust impacted by external influences, such as the USMC or the US government. Feelings of trust highlighted the HSTCA domains of confidence, fidelity, honesty, communication, competence, confidentiality, systems trust, and fairness.

Although assigned to a specific HSTCA domain, there is great overlap between domains in that one illustrative quote, or Veteran experience, can relate to multiple domains. These findings suggest that trust in the VHA is complex and shaped by a nuanced set of factors, including potentially modifiable factors such as access to VHA care, experiences with submitting claims for conditions perceived to be related to toxicant exposure, and the perceived transparency of the USMC, VHA, and US government.

Although VA-wide surveys show high levels of trust in the organization attributed to ease, effectiveness, and emotion,²¹ our findings from this study along with our team's prior work suggest that the needs of CL Veterans may be unique and require tailored efforts to address trust.²⁸ Beyond this, this investigation into the lived experiences of some of these Veterans revealed a potential need to rebuild trust in this subpopulation of Veterans exposed at CL. Veterans may have healthcare service options available to them, however, evidence suggests that lack of trust can decrease use of these services.¹ A recent study found that Veterans with higher trust in VHA physicians were more than twice as likely to choose VHA over non-VHA care.²⁹ Because the VHA has the resources and knowledge to treat and compensate this vulnerable population, restoring trust with CL Veterans, such as those who fit in our lack of trust theme, may be essential for Veterans to choose VHA care over other providers.³⁰ These restorative efforts may span multiple levels of the VA from VHA provider encounters to claims legislation.

Future research should include identifying measures to assess trust in this population, building on the VHA's existing system-wide initiatives to monitor and increase trust.²¹ Objective measurement of trust among CL Veterans will be critical to establish the key drivers of trust, including when trust was established and/or broken and the influence of perceptions and experiences with the larger VA, USMC, and US government on CL Veteran trust in the VHA. This measurement is also critical in establishing strategies to restore trust in the CL community. We propose that future efforts may be most beneficial if they: (1) leverage community-engaged research principles in program planning and policymaking; (2) lean into Veterans' camaraderie; and (3) understand how relationships with communities experiencing past environmental exposures have unfolded throughout history.

First, our findings further develop a foundation for leveraging community-engaged research principles to better understand and address trust in the VHA among CL Veterans. Community-engaged research structures can empower Veterans' decision-making and active engagement through community advisory boards (CABs) or steering committees, inviting them as partners in trust-related initiatives. For example, the VHA may increase efforts to engage CL Veterans, their family members, VHA physicians and employees, researchers, and policymakers when strengthening the disability claims process or creating other resources to help

Veterans navigate associated legislation. In a recent editorial, Taylor Henson, a retired Marine and public health worker, explained how a disconnect between public health and Veteran language can inadvertently drive Veterans further away from the public health initiatives designed to help them.³¹ This disconnect stems from a lack of understanding of community culture (language, experience, tradition, values, etc.) and can make Veterans question the efficacy of suggested programs as they may appear uninformed even if they are rigorous or proven.³¹

In characterizing Veterans' feelings of trust alongside an established framework, we were able to identify where these disconnects might occur (e.g., Table 2). Frameworks, such as the HSTCA, may also contribute toward standardized measures of trust that streamline the way we evaluate program effectiveness while integrating Veterans' perceptions, values, and lived experiences. This way, the VHA can focus any future efforts to restore trust on these eight domains as they relate to their specific systems and policies, showcasing our operational definitions as a novel contribution to the field.

Second, leaning into Veterans' camaraderie, or trust among other military personnel, is essential when considering trust with this community. Experiences of trusted, peer-level individuals (reaching horizontally, rather than vertically to institutions in power) can impact trust related to healthcare because of the relatability and inherent bond Veterans feel with one another.² In one study, 82% of survey respondents reported that their perceptions of the VA were shaped by the experiences of someone they knew personally,²² while Veterans who valued camaraderie when choosing a healthcare system were nearly 1.5 times more likely to choose the VA over other healthcare systems.²⁹ These findings showcase the need to highlight camaraderie as part of the appeal for choosing VHA hospitals over other options, and to amplify voices of CL Veterans who had positive experiences with the VHA, such as those in our trust theme, as champions for VHA services and programs. Current VA programs that already involve Veteran-to-Veteran support should be further emphasized to CL Veterans, including the use of Veteran Service Officers,³² or tailored to CL Veterans using our findings on trust as a blueprint, such as the creation of peer support specialists³³ specifically for those with military environmental exposures.

Third, it can be valuable to reflect on how other environmental disasters have impacted relationships with communities throughout history. Successes and pitfalls surrounding the handling of environmental health disasters will undoubtedly offer lessons that can be applied to address trust among CL Veterans. These events include military-related environmental disasters, such as exposures to Agent Orange,³⁴ burn pits in Iraq, Afghanistan, and other areas of Southwest Asia,³⁵ and Gulf War related illnesses;³⁶ as well as non-military related environmental responses, like the Minamata Bay

methylmercury leak which caused severe, long-term health effects for communities in Minamata Bay, Japan.³⁷ Although the specific needs and contexts of CL Veterans should still be taken into consideration, leaning on lessons learned from these examples may increase program effectiveness and Veteran confidence that programs were thoughtfully planned, ultimately increasing trust. For example, lessons learned from Minamata Bay include the importance of accountability, acknowledgement of community experiences, expansion of healthcare relief and compensation, and investment in continued research.³⁸ That being said, these findings on trust within CL Veterans also offer lessons for future Veteran communities who face military environmental toxicants as laid out by the tailored HSTCA framework.

This study has important limitations, including non-response bias where those with extreme views may be more likely to volunteer as an interview participant.³⁹ Therefore, our study findings may not be generalizable to the broader CL Veteran population or to all Veterans exposed to military toxicants. Besides those recruited from Facebook, contact information for our interview participants was obtained through the VA Corporate Data Warehouse and only included Veterans who used VA care because we did not have a way to contact CL Veterans who received healthcare elsewhere. Similarly, this study may be impacted by sample bias as those with higher trust in the VA may have agreed to participate with our team, though more experiences describing a lack of trust were shared than trust. Future work in this area should consider how to reach—and factor in the perspectives of—CL Veterans who choose to seek healthcare outside of the VA. Finally, these data were self-reported and subject to recall bias.

In terms of the use of the HSTCA framework, there is high overlap between domains, and we are only able to assess trust at one timepoint.⁴⁰ Future studies may benefit from the evaluation of changes in trust across military service (pre- to post-service).

CONCLUSIONS

This qualitative study of trust among CL Veterans exposed to military environmental toxicants makes a novel contribution in shedding light on trust in the VA based on positive experiences with receiving healthcare and disability approval. On the other hand, CL Veterans also discussed a lack of trust shaped by dissatisfaction with VHA healthcare, the complexity of the VA's claims processes, and a perceived lack of transparency about the exposure. Similarly, Veterans shared that external influencers, such as the US government and USMC, also impacted trust. Our findings begin to fill an important gap in knowledge on trust in the VHA following military environmental exposure at CL and can be used to inform future policy and community-engagement efforts to

optimize the health and healthcare of Veterans with such exposures.

Because trust is critical for the use of healthcare services, it is necessary for the VHA to rebuild trust with Veterans impacted by military environmental exposures, so Veterans seek the treatment and compensation they need. Lessons learned may also inform the way that the US government, armed forces, and VA approach communities if future military environmental exposure occurs. Next steps may include finding a more holistic and standardized way of measuring trust to provide meaningful comparison of trust between studies and frameworks. Future research may aim to answer these key questions to inform policy changes: (1) what are the reasons for continued VHA use regardless of Veteran trust? (2) how can the VHA rebuild trust with Veterans? (3) how can we shift Veteran perspectives towards receiving help from the VHA using new legislation and communication? Next steps should also include actively engaging Veterans to pinpoint where modifications can be made specific to the CL community, outlining actionable strategies to improve trust, including potential revisions to the claims process, and acknowledgement that trust in the VHA can be influenced by factors beyond the VHA, such as the USMC and the US government.

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Data Availability The datasets generated during and/or analyzed during the current study are not publicly available due to privacy restrictions but are available from the corresponding author on reasonable request.

Declarations

Ethics Approval and Consent to Participate Informed consent was obtained prior to interviews for all participants.

Conflict of Interest All authors declare no conflicts of interest other than funding to the institution.

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